

Answer of Question no. 1

In quantum physics, a particle is represented by a wave packet rather than by a single monochromatic wave.

Two important velocities associated with a wave are phase velocity and group velocity. Phase velocity represents the speed at which particular phase of the wave propagates, whereas group velocity represents the speed at which the wave packet or group of waves travels.

Answer of Question no. 2

Interference is a fundamental phenomenon of wave optics in which two or more coherent light waves superpose to produce a resultant intensity distribution containing bright and dark regions.

Consider two waves

$$y_1 = a_1 \sin \omega t$$

and

$$y_2 = a_2 \sin(\omega t + \phi)$$

The resultant intensity is

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$$

Answer of Question no. 3

A LASER stands for Light Amplification by Stimulated Emission of Radiation. A Laser is a device that produces highly intense, coherent, monochromatic and directional electromagnetic radiation.

The basic principle of laser operation is based on the interaction between atoms and electromagnetic radiation.

There are three important processes:

- ① Absorption
- ② Spontaneous emission
- ③ Stimulated emission

Answer of Question no. 4

The electric field is the region around an electric charge in which another electric charge experiences an electric force.

The electric field is represented by the symbol E .

If a test q experiences a force F , then electric field is defined as

$$E = \frac{F}{q}$$

The SI Unit of electric field is N/C or V/m .

Answer of Question no. 5

A crystal is a solid in which atoms, ions or molecules are arranged in a regular and periodic pattern. The study of crystal structure is important in solid-state physics because the arrangement of atoms strongly influences electrical, thermal, optical and mechanical properties.

Important parameters

A Unit cell is described using lattice constraints

and the angles a, b, c
 α, β, γ